Vansh Gupta

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CodeChef **4-Star** (max 1856) | Codeforces **Pupil** (max 1340) | Bhopal, India

Professional Summary

Third-year B.Tech student at IIIT Bhopal (CGPA: **8.49/10**) with 2+ years designing, training, and evaluating neural network architectures across **Computer Vision**, **NLP**, and **LLM** applications — building production-quality AI systems that ship to real users. Hands-on experience with **supervised learning**, **pattern recognition**, **text classification**, and **information extraction** using PyTorch, TensorFlow, and Scikit-learn, with deep understanding of the mathematics behind AI, transformers, and neural networks. Strong mathematical foundation validated at a national level (**98.16 percentile JEE Main**; **AIR 27**, national mathematics finals co-organized by IIT Bombay & IIT Madras) — directly applicable to the mathematical rigour required for AI model design and optimization. Currently applying this foundation as an ML Intern at Moleculyst, building and evaluating production computer vision and NLP systems in a fast-paced R&D environment.

Education

IIIT Bhopal — **Indian Institute of Information Technology** **Bhopal, Madhya Pradesh, India**
B.Tech in Information Technology (CS & AI/ML Curriculum) | *CGPA: **8.49 / 10*** *Aug 2024 – May 2028*

- **Coursework:** Machine Learning, Artificial Intelligence, Computer Vision, Natural Language Processing, Neural Networks, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Discrete Mathematics

Technical Skills

AI/ML Fundamentals: Supervised Learning, Unsupervised Learning, Neural Network Design & Training, Model Evaluation & Optimization, Pattern Recognition, Data Preprocessing, Feature Engineering

Computer Vision & NLP: Computer Vision (VLMs, OpenCV, Image Grounding), Natural Language Processing, Text Classification, Intent Classification, Information Extraction, Conversational Interfaces, Sentiment Analysis

Neural Networks & Frameworks: PyTorch, TensorFlow, Scikit-learn, LSTM (Recurrent Networks), XGBoost, Transformer Architectures, SHAP (Model Interpretability), Reinforcement Learning

Mathematics of AI: Linear Algebra, Calculus, Probability & Statistics, Mathematical Optimization, Stochastic Modeling — *nationally validated: 98.16 percentile JEE Main | AIR 27, National Maths Finals (IIT Bombay & IIT Madras)*

LLM & Generative AI: Large Language Models (Claude, GPT-4, Gemini, LLaMA 3.1), RAG Pipelines, Prompt Engineering, LangChain, LangGraph, Hugging Face, Ollama

Production & Dev Tools: Python, FastAPI, Docker, Git / GitHub (Version Control), Google Cloud Run, Vertex AI, AWS, REST APIs, Production-Quality Code

Experience

ML Intern **Feb 2026 – Present**
Moleculyst *Remote, India*

- **Neural Network Training:** Built and optimized Video LLM inference pipelines in Python, designing and integrating **transformer-based** vision-language neural network architectures for **computer vision** and **NLP** tasks — delivering production-quality AI software for structured understanding across complex video datasets.
- **Model Evaluation & Optimization:** Designed structured evaluation frameworks defining accuracy, quality, and reliability metrics across model versions, enabling systematic **model performance refinement** and reproducible benchmarking without manual intervention.
- **Research Translation:** Translated AI/ML research papers on quantitative methods, mathematical optimization, and statistical modeling into working production code — demonstrating the ability to understand and apply the **mathematics behind AI** and transformer architectures in an R&D environment.
- **Data Pipeline:** Supported **data collection and preprocessing** for multimodal datasets, performing feature engineering and collaborating with cross-functional teams to define clean, model-ready input pipelines for neural network training.

Projects

Moleculyst Studio | *Python, Vision-Language Models (VLMs), OpenCV, Gemini API, Docker* **2025**

- Built a production AI workspace integrating **computer vision** and **NLP** via Vision-Language Models (VLMs) for **image grounding and pattern recognition**, composing targeted editing operators and an automated **LLM-as-judge evaluation loop** — demonstrating hands-on design and evaluation of neural network-based CV and language systems.
- Implemented production-quality model **performance monitoring** by streaming structured reasoning traces, tool call outputs, and per-step quality scores into a unified view — enabling iterative optimization and full observability of neural network outputs at runtime.

PharmAI | *Python, LSTM, XGBoost, SHAP, Scikit-learn, PyTorch* **2025**

- Designed and trained **LSTM (recurrent neural network)** and XGBoost architectures for **supervised learning** on pharmaceutical batch quality prediction — implementing the full ML pipeline from **data preprocessing and feature engineering** to model evaluation, achieving **>90% accuracy** on held-out validation sets.

- Built a SHAP-driven **pattern recognition and anomaly detection** framework for early failure identification, quantifying model interpretability by surfacing top contributing variables per flagged prediction — demonstrating production-grade model evaluation in a regulated domain.

Triage AI | *Python, FastAPI, OpenEnv, LangChain, Docker, LLaMA 3.1* **2025**

- Architected a reinforcement learning environment for **NLP-driven text classification** and intelligent routing, training **LLaMA 3.1 (8B)** — an 8-billion parameter **Transformer architecture** — with multi-level prompt engineering chains and achieving **0.71 / 1.0** classification accuracy.
- Designed a structured **model evaluation framework** with reward-based accuracy metrics and containerized the full system with Docker, demonstrating production-quality software engineering and version-controlled deployment of trained NLP models.

Equitas AI | *Python, LangChain, LangGraph, FastAPI, Gemini API, Vertex AI* **2025**

- Designed a 5-agent NLP pipeline for automated **information extraction** and bias analysis, implementing SHAP explainability matrices with Gemini-driven **conversational interface** output — achieving Disparate Impact Ratio (DIR) $\geq$ **0.80** across all evaluated benchmarks.
- Deployed production-quality AI software on Google Cloud Run via Vertex AI using FastAPI — demonstrating robust, scalable integration of NLP models into a real-world client-facing system with version control and zero credential overhead.

Achievements

- **SRMC Mathematics Competition — AIR 27 (National Finals):** Ranked **27th nationally** in the mathematics tournament jointly organized by **IIT Bombay & IIT Madras** — demonstrating the deep mathematical foundation required for AI model design, transformers, and optimization.
- **JEE Main 2024 — 98.16 percentile:** Top 1% nationally out of 1.2M+ candidates; qualified for JEE Advanced — validating strong foundations in mathematics, physics, and analytical reasoning.
- **Meta × PyTorch OpenEnv Hackathon — Grand Finalist** (Bangalore, 2025): Top-tier ranking out of **52,000+** global participants for building a production reinforcement learning and NLP classification system.
- **OpenAI × Outskill AI Builders Hackathon:** Shortlisted in **Top 1,000** developers nationwide for designing and shipping a production-grade live AI product.
- **CodeBidz Hackathon — 1st Place:** Won as a solo developer executing the full product lifecycle independently.